SOCIAL MEDIA ANALYTICS

50 Important Viva Questions & Answers

Complete Formula Sheet for Units 2, 4 and 5

Prepared for quick exam revision. Includes 10 short question-and-answer pairs for each of the five units, followed by formulas collected from the uploaded unit notes and clearly labelled additional standard advertising formulas.

Viva tip: Start with the definition, then give one short example. For numerical questions, state the formula before substituting values.

UNIT 1: FOUNDATIONS OF SOCIAL MEDIA ANALYTICS & DIGITAL ECOSYSTEM

1. What is digital marketing?

Answer: Digital marketing is the promotion of products or services through digital channels such as websites, search engines, email and social media.

2. What is social media analytics?

Answer: Social media analytics is the process of collecting and analyzing social media data to understand audience behaviour and measure performance.

3. How has digital marketing evolved?

Answer: It evolved from early customer databases and email/web marketing to social media interaction, data-driven advertising and AI-based personalization.

4. What is Web 1.0?

Answer: Web 1.0 is the early, mostly static and read-only web where users mainly consumed information rather than interacting with it.

5. What is user-generated content (UGC)?

Answer: UGC is content created by users, such as reviews, photos, videos and comments.

6. Name five social media platforms.

Answer: Facebook, Instagram, YouTube, LinkedIn and X are five examples.

7. What is the difference between organic and paid social media?

Answer: Organic social media uses unpaid posts and distribution. Paid social media uses advertising spend to reach selected audiences.

8. What are owned, earned and paid media?

Answer: Owned media is controlled by the brand, such as its website. Earned media is unpaid publicity or mentions. Paid media is exposure purchased through advertising.

9. What is influencer marketing?

Answer: Influencer marketing is when a brand collaborates with content creators to promote a product, service or message to their audience.

10. What is omnichannel marketing?

Answer: Omnichannel marketing provides a connected and consistent customer experience across channels such as websites, apps, email and social media.

UNIT 2: METRICS, KPIs & MEASUREMENT FRAMEWORKS

1. **What is a KPI?**
   Answer: A Key Performance Indicator (KPI) is a measurable value used to evaluate progress toward a campaign or business objective.

2. **What is reach?**
   Answer: Reach is the number of unique users who see a piece of content.

3. **What are impressions?**
   Answer: Impressions are the total number of times content is displayed, including repeated displays to the same person.

4. **What is the difference between reach and impressions?**
   Answer: Reach counts unique people, while impressions count total displays. If 100 people each see a post twice, reach is 100 and impressions are 200.

5. **How is engagement rate calculated?**
   Answer: Engagement rate is commonly calculated as total engagements divided by reach, multiplied by 100.

6. **What is click-through rate (CTR)?**
   Answer: CTR is the percentage of impressions that result in clicks. Formula: clicks divided by impressions, multiplied by 100.

7. **What is a lead?**
   Answer: A lead is a person who has shown interest in a product or service, for example by submitting a contact form.

8. **What is conversion rate?**
   Answer: Conversion rate is the percentage of a specified group—such as visitors or ad clickers—who complete the desired action.

9. **What are CAC and CLV?**
   Answer: Customer Acquisition Cost (CAC) is the average sales and marketing cost to acquire one new customer. Customer Lifetime Value (CLV) estimates the value a customer generates over the relationship.

10. **What are amplification rate and applause rate?**
    Answer: Amplification rate measures sharing relative to followers. Applause rate measures likes relative to followers.

UNIT 3: ANALYTICS TOOLS, DATA MANAGEMENT & REPORTING

1. **What is Google Analytics?**
   Answer: Google Analytics is a web analytics tool used to measure website traffic, user behaviour, traffic sources and conversions.
2. **Name five metrics tracked in Google Analytics.**
   Answer: Examples include users, sessions, page views, bounce rate and conversion rate.
3. **What is Meta Business Suite?**
   Answer: Meta Business Suite is a centralized platform for managing Facebook and Instagram business activities, including publishing, messages, advertising and analytics.
4. **What is Hootsuite?**
   Answer: Hootsuite is a social media management platform used to manage multiple social accounts, schedule content, monitor activity and prepare reports.
5. **What is the difference between Meta Business Suite and Hootsuite?**
   Answer: Meta Business Suite focuses on Meta platforms, especially Facebook and Instagram. Hootsuite supports management across multiple social networks.
6. **What social media data can be collected?**
   Answer: Common data includes likes, comments, shares, views, reach, impressions, audience details, clicks and advertising performance.
7. **What are platform insights?**
   Answer: Platform insights are native analytics dashboards that report measures such as reach, engagement, followers, video views and audience demographics.
8. **What is an API?**
   Answer: API stands for Application Programming Interface. It allows software applications to communicate and exchange data.
9. **Name some types of APIs used for data access.**
   Answer: Examples discussed in the notes include REST APIs, streaming APIs and marketing APIs.
10. **What is the purpose of dashboards and reports?**
    Answer: Dashboards and reports organize metrics so users can monitor performance, identify trends, compare results and make decisions.

UNIT 4: SOCIAL MEDIA DATA ANALYSIS & CONSUMER INSIGHTS

1. **What is data interpretation?**

   Answer: Data interpretation means converting collected social media data into meaningful information that supports decisions.

2. **Name some social media data analysis techniques.**

   Answer: Examples include descriptive, comparative, trend, sentiment, conversion, segmentation, correlation, funnel and attribution analysis.

3. **What is descriptive analysis?**

   Answer: Descriptive analysis summarizes what has already happened using measures such as reach, impressions, likes, comments, shares and clicks.

4. **What is trend analysis?**

   Answer: Trend analysis examines changes and patterns over time, such as growth in followers, engagement, traffic or sales.

5. **What is engagement pattern analysis?**

   Answer: It studies how audiences interact with content across platforms, content types, audience groups and posting times.

6. **What is sentiment analysis?**

   Answer: Sentiment analysis classifies text or opinions, commonly as positive, negative or neutral, to understand audience attitudes.

7. **Give examples of positive, negative and neutral sentiment.**

   Answer: “Excellent product!” is positive; “The service is poor” is negative; “The product comes in three sizes” is neutral.

8. **What is segmentation analysis?**

   Answer: Segmentation analysis divides an audience into groups based on factors such as age, location, interests, buying behaviour or engagement.

9. **What is funnel analysis?**

   Answer: Funnel analysis tracks users through stages such as awareness, interest, consideration, conversion and retention to identify where users drop out.

10. **What is attribution analysis?**

    Answer: Attribution analysis determines how credit for a conversion is assigned to marketing touchpoints. Models include first-click, last-click, linear, time-decay and position-based.

UNIT 5: STRATEGY, OPTIMIZATION, ETHICS & CAPSTONE PROJECTS

1. **What is data-driven campaign planning?**
   Answer: It is the use of analytics, customer data and insights to plan, run, monitor and improve marketing campaigns.
2. **How does data-driven marketing differ from traditional planning?**
   Answer: Traditional planning may rely more on assumptions and broad targeting. Data-driven planning uses evidence, audience segments, KPIs and ongoing optimization.
3. **What are the main steps in campaign planning?**
   Answer: Set objectives, define the audience, select platforms, review past data, plan content and timing, set the budget and KPIs, launch, monitor and optimize.
4. **What is A/B testing?**
   Answer: A/B testing compares two versions of content or an ad. Version A is the control and version B changes an element, such as a headline, to see which performs better.
5. **Why is A/B testing important?**
   Answer: It helps identify better-performing content, improve engagement and conversions, reduce wasted spend and support evidence-based decisions.
6. **What is campaign optimization?**
   Answer: Campaign optimization means adjusting targeting, creative, copy, calls to action, budget, timing or landing pages to improve performance.
7. **Name categories of social media KPIs.**
   Answer: Categories include awareness, engagement, traffic, lead generation, conversion, customer, advertising, financial and brand metrics.
8. **How is ROI calculated?**
   Answer: ROI is calculated as (gain minus investment cost) divided by investment cost, multiplied by 100.
9. **What are ethics in social media analytics?**
   Answer: Ethics means handling data responsibly, respecting privacy, being transparent and avoiding misleading or unfair practices.
10. **What is a capstone project?**
    Answer: A capstone project is a practical project that applies learned concepts to a real or simulated problem through data collection, analysis, interpretation and recommendations.

FORMULA SHEET

Key calculations for social media analytics

Note: Some metrics have different denominator choices depending on the exact funnel stage or platform definition. Use the denominator specified in the question or in the relevant unit example.

UNIT 2: METRICS, KPIs & MEASUREMENT FRAMEWORKS

Metric	Formula / Definition
Reach	Number of unique users who viewed the content.
Impressions	Total number of times content was displayed.
Engagement Rate	(Total Engagements ÷ Total Reach) × 100
Click-Through Rate (CTR)	(Clicks ÷ Impressions) × 100
Lead Generation Rate	(Total Leads ÷ Total Visitors) × 100
Conversion Rate	(Conversions ÷ relevant denominator) × 100. Use visitors, clicks or leads as the denominator according to the question and the definition in the notes.
Bounce Rate	(Single-Page Sessions ÷ Total Sessions) × 100
Customer Acquisition Cost (CAC)	Total Sales and Marketing Expenses ÷ Number of New Customers Acquired
Customer Lifetime Value (CLV/LTV)	Average Purchase Value × Purchase Frequency × Customer Lifespan
Amplification Rate	(Total Shares ÷ Total Followers) × 100
Applause Rate	(Total Likes ÷ Total Followers) × 100

UNIT 4: SOCIAL MEDIA DATA ANALYSIS & CONSUMER INSIGHTS

Metric	Formula / Definition
Growth Rate (%)	((New Value – Old Value) ÷ Old Value) × 100
Engagement Rate (%)	((Likes + Comments + Shares) ÷ Reach) × 100. If the task includes saves or other engagements, include those in total engagements.
Conversion Rate (%)	(Conversions ÷ Clicks) × 100, for the click-to-conversion example in the notes.

UNIT 5: STRATEGY, OPTIMIZATION & PERFORMANCE METRICS

Metric	Formula / Definition
Engagement Rate	(Total Engagements ÷ Reach) × 100
Click-Through Rate (CTR)	(Clicks ÷ Impressions) × 100
Conversion Rate	(Conversions ÷ Clicks) × 100, as shown in the Unit 5 example.
Cost Per Click (CPC)	Advertising Cost ÷ Number of Clicks
Cost Per Lead (CPL)	Advertising Spend ÷ Number of Leads
Return on Investment (ROI)	((Gain – Investment Cost) ÷ Investment Cost) × 100
Customer Lifetime Value (CLV)	Average Purchase Value × Purchase Frequency × Customer Lifespan
Monthly Average Budget	Annual Budget ÷ 12
Budget Allocation (%)	(Budget Allocated to a Platform ÷ Total Budget) × 100

ADDITIONAL STANDARD ADVERTISING FORMULAS

Metric	Formula / Definition
Cost Per 1,000 Impressions (CPM)	(Advertising Cost ÷ Impressions) × 1,000
Cost Per Acquisition (CPA)	Advertising Cost ÷ Number of Acquisitions
Return on Ad Spend (ROAS)	Revenue Attributed to Ads ÷ Advertising Spend

Worked examples

Growth rate

Clicks increased from 2,000 to 2,500: ((2,500 – 2,000) ÷ 2,000) × 100 = 25%.

Engagement rate

800 likes + 100 comments + 100 shares = 1,000 engagements. If reach is 10,000, rate = (1,000 ÷ 10,000) × 100 = 10%.

CTR

50 clicks from 1,000 impressions: (50 ÷ 1,000) × 100 = 5%.

CAC

■1,00,000 in sales and marketing expenses for 100 new customers: ■1,00,000 ÷ 100 = ■1,000 per customer.

CLV

Average purchase value ■2,000 × 5 purchases per year × 3 years = ■30,000.

ROI

If gain is ■15,000 and investment cost is ■10,000: ((■15,000 – ■10,000) ÷ ■10,000) × 100 = 50%.

CPC

■20,000 advertising cost ÷ 5,000 clicks = ■4 per click.

Source note: The 50 viva answers follow the five-unit study set previously summarized from the uploaded notes. The main formulas are drawn from the retrieved material for Units 2, 4 and 5. CPM, CPA and ROAS are included as additional standard definitions because they are named as performance metrics in the notes, although their formulas were not all explicitly visible in the retrieved excerpts. No separate calculation formula was identified in the retrieved Unit 3 material.